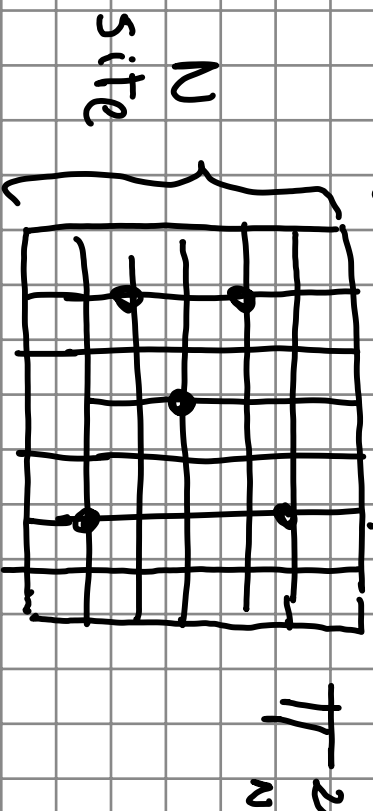


Introduction to interacting particle systems (IPS)

Ty model of non equilibrium on a lattice (π^d_N)



$\eta(x)$ is the occupation variable at x , $\underline{\eta} = (\eta(x))_{x \in \mathbb{Z}^N}$ configuration that evolves into

$\underline{\eta}(t) = \{\eta_t(x)\}_x$ with particles performing R.W.

and rules (hardware) encoded in a Markov generator

These dynamics allows to derive laws (hydrodynamics) for macroscopic observables,

e.g.
$$\left[\pi u(n, du) = \frac{1}{N^d} \sum_x \eta(x) \delta_x(du) \right] \quad \text{empirical measure}$$

We can prove $\partial_t \rho = \partial_x (D(\rho) \partial_x \rho)$ as LLN.

Microscopic conservation law

$$\left[\eta_t(x) - \eta_0(x) + \nabla \cdot \underbrace{J_t(x)}_{\substack{:= \\ \sum_{y \sim x} J_t(x,y)}} = 0, \quad \forall x \right]$$

$J_t(x,y)$ is the net flow of particles that crossed (x,y) in $[0,t]$.

Programe

-) Chap. 4 of KL
-) What was known + discrete exterior calculus
-) my results.